

APPLIED MATHEMATICS (CLASS 12)

 Top 50 Most Repeated MCQs by Thunderstudy

These MCQs are curated from CBSE previous years' papers and recurring concepts (including Q.P. Code 465). Perfect for board revision.

MOD, MATRICES & ALGEBRA

1. Smallest positive integer congruent to 282 (mod 11) is

- A) 3 B) 7 C) 9 D) 17

2. If $AB = A$ and $BA = B$, then B^2 equals:

- A) $2A$ B) O C) $2I$ D) B

3. If A is invertible, then which is NOT true?

- A) $(A^{-1})^{-1} = A$ B) $(A^T)^{-1} = (A^{-1})^T$ C) $(AB)^{-1} = B^{-1}A^{-1}$ D) $\det(A^{-1}) = \det(A)$

4. System of equations is consistent if determinant $\neq ?$

- A) 0 B) 1 C) -1 D) 2

5. Inverse of matrix exists when:

- A) determinant = 0 B) determinant $\neq 0$ C) matrix symmetric D) matrix square only

FUNCTIONS & CALCULUS

6. a^x is increasing on $\mathbb{R}$ when:

- A) $a > 0$ B) $a > 1$ C) $a < 0$ D) $0 < a < 1$
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7. Function $f(x) = x^3$ has:

- A) max only B) min only C) both D) neither
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8. If $y = \log(x)$, then dy/dx equals:

- A) $1/x$ B) e^x C) x D) none
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9. Order of differential equation =

- A) highest derivative power B) highest derivative order C) degree D) constant
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10. Marginal Cost (MC) & Average Cost (AC) relation:

- A) $d/dx(AC) = 1/x(MC - AC)$ B) $x(MC - AC)$ C) $MC + AC$ D) none
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PROBABILITY & DISTRIBUTION

11. Expectation of heads when coin tossed twice:

- A) 1 B) $\frac{1}{2}$ C) $\frac{1}{4}$ D) 2
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12. Mean of t-distribution is:

- A) 0 B) 1 C) 2 D) undefined
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13. Variance = SD^2 ; if $SD = \sqrt{2}$, variance equals?

- A) 8 B) 5 C) 2 D) 1
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14. Binomial distribution mean formula is:

- A) np B) npq C) $\sqrt{npq}$ D) n/p
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15. If mean = 12, SD = 2 for a binomial distribution $\rightarrow p = ?$

- A) $\frac{5}{6}$ B) $\frac{1}{6}$ C) $\frac{1}{3}$ D) $\frac{2}{3}$
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16. Variance of Poisson distribution equals:

- A) mean B) SD C) median D) mode
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17. Normal distribution curve is symmetric about:

- A) variance B) mean C) SD D) covariance
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LINEAR PROGRAMMING

18. Optimal value (maximum/minimum) occurs at:

- A) interior point B) corner point C) origin only D) midpoint
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19. If objective function is same at two vertices:

- A) 1 solution B) 2 solutions C) finite solutions D) infinite solutions
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20. Inequality $ax + by \leq c$ represents:

- A) full plane B) half-plane C) line only D) point
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FINANCIAL MATHEMATICS

21. EMI (Flat rate) formula calculation uses:

- A) simple interest B) compound interest C) discounting D) annuity
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22. Present value of perpetuity =

- A) $\frac{R}{i}$ B) Ri C) $R+i$ D) $R-i$
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23. Effective rate > nominal rate when interest is:

- A) compounded B) simple interest C) discounting D) none

24. CAGR formula =

- A) $[(FV/PV)^{(1/n)}] - 1$ B) $FV - PV$ C) FV/PV D) none

25. Book value of an asset decreases due to:

- A) depreciation B) appreciation C) tax D) inflation

STATISTICS & TIME SERIES

26. Seasonal variation is caused by:

- A) climate/customs B) random events C) long term trend D) business cycle

27. Moving average method removes:

- A) seasonal variation B) random variation C) trend D) none

28. Degrees of freedom for two independent samples:

- A) $n_1 + n_2 - 2$ B) $n_1 - n_2$ C) $n_1 + n_2$ D) $n_1 - 2$

LOGIC & INFERENCE STATISTICS

29. If $|t\text{-calculated}| < t\text{-critical value}$:

- A) reject H_0 B) accept H_0 C) inconclusive D) none

30. In Assertion-Reason, if Assertion is true but Reason is false:

- A) Option A B) Option B C) Option C D) Option D

APPLICATION & CONCEPTUAL FILLERS

31. Upstream speed calculation when upstream time is double downstream time:
Concept of **Relative Speed**.

32. Basic population growth differential equation is in the form: $\frac{dP}{dt} = kP$.

33. The derivative of a demand function or total revenue leads to: **Marginal Revenue (MR)**.

34. Profit is maximized at a production level where: **MR = MC**.

35. Average cost is at its minimum when: **MC = AC**.

MIXED HIGH-FREQUENCY CONCEPTS

36. Expected value $E(X)$ formula = $\sum x_i p_i$.

37. Variance is always **greater than or equal to 0**.

38. Probability of any event always lies **between 0 and 1**.

39. A Determinant **changes sign** when any two rows or columns are swapped.

40. The determinant of an Identity matrix is always **1**.

41. A linear inequality in two variables always represents a **half-plane**.

42. Feasible region in LPP is the **intersection area** of all constraints.

43. The Normal distribution curve is always **bell-shaped**.

44. In a perfectly Normal Distribution: **Mean = Median = Mode**.

45. Poisson distribution is typically used for **rare events**.

46. In a Binomial distribution, all trials must be **independent**.

47. A function is increasing if its first derivative is **positive (>0)**.

48. A function is decreasing if its first derivative is **negative (<0)**.

49. Continuous compounding uses the **exponential growth** function (e^{rt}).

50. In LPP, the objective function is always **optimized at the vertices** of the feasible region.



ANSWER KEY

1: C (9)	2: D (B)	3: D	4: A (0)	5: B ($\neq 0$)
6: B (>1)	7: D	8: A ($1/x$)	9: B	10: A
11: A (1)	12: A (0)	13: C (2)	14: A (np)	15: C ($1/3$)
16: A (mean)	17: B (mean)	18: B	19: D	20: B
21: A (SI)	22: A (R/i)	23: A	24: A	25: A
26: A	27: A	28: A	29: B	30: C
31: Rel. Spd	32: kP	33: MR	34: MR=MC	35: MC=AC
36: Σxp	37: True	38: True	39: True	40: True
41: True	42: True	43: True	44: True	45: True
46: True	47: True	48: True	49: True	50: True

Top 50 MCQs with answers by Thunderstudy | Do not remove text